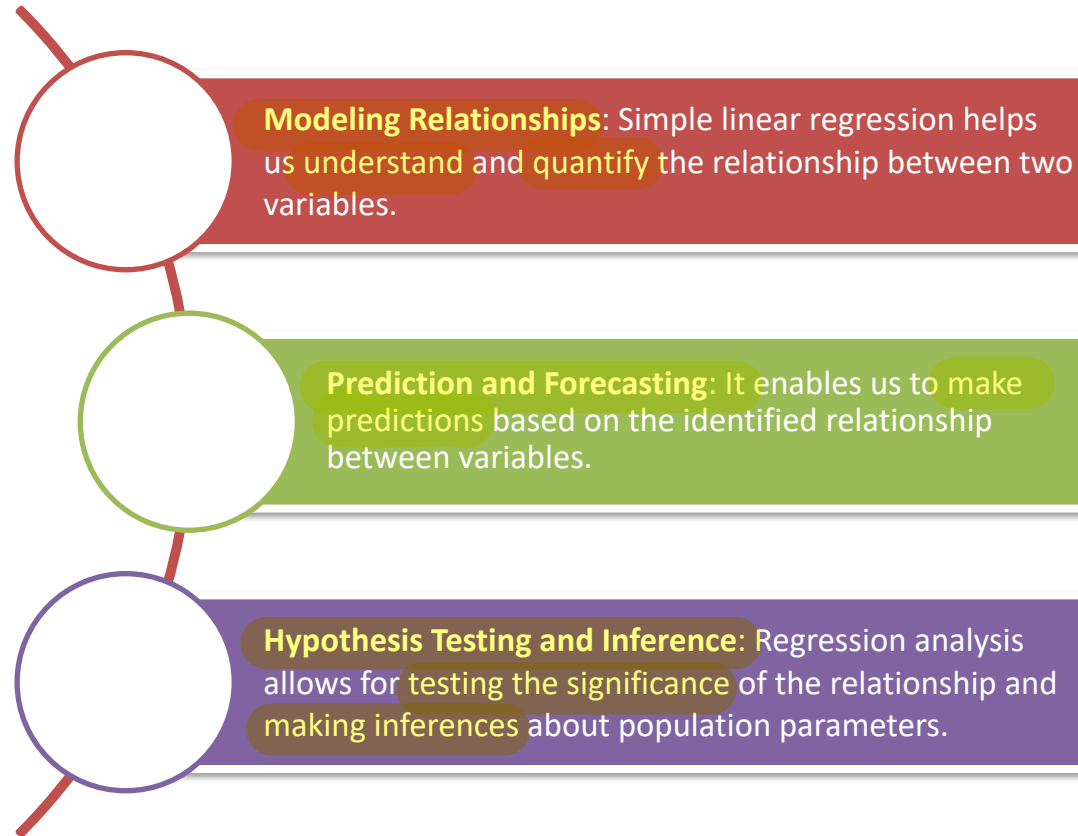


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Stat 3113: Lesson 35 Regression Examples

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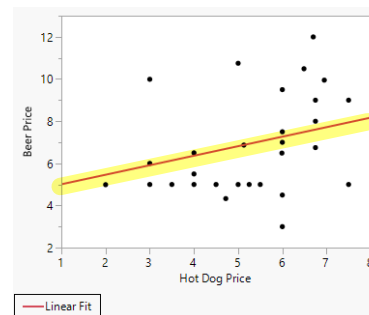
Motivations



MORE REGRESSION EXAMPLES

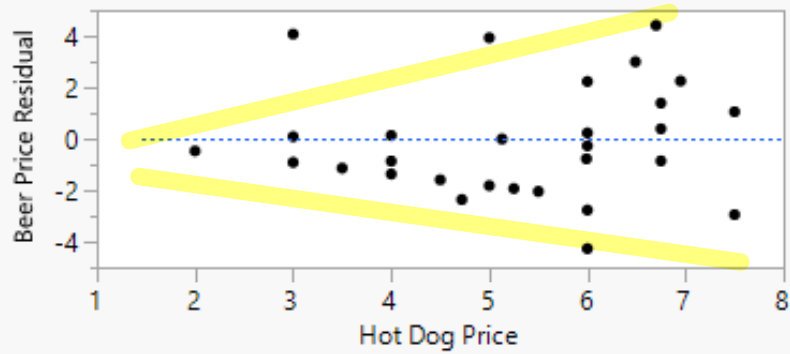
Example 1: Data from 2022

- x =Hot dog price at major league ballpark and y =Beer price at major league ballpark
- Suppose we wish to see if the hot dog price at a park can be used to predict the beer price.

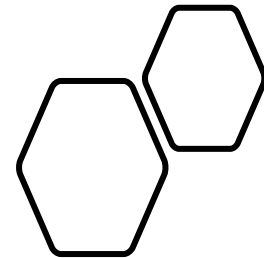
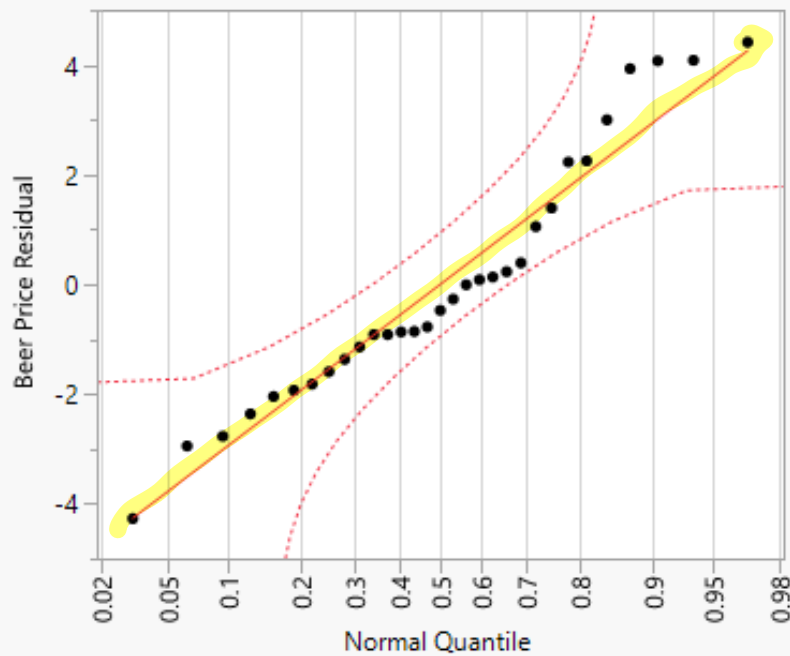


Linear Fit				
Beer Price = 4.5594683 + 0.4514183*Hot Dog Price				
Summary of Fit				
R Square				0.087048
R Square Adj				0.055567
Root Mean Square Error				2.287085
Mean of Response				6.874516
Observations (or Sum Wgts)				31
Lack Of Fit				
Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	14.46359	14.4636	2.7651
Error	29	151.69198	5.2308	Prob > F
C. Total	30	166.15557		0.1071

Residual by X Plot



Residual Normal Quantile Plot



a. What is the equation of the least-squares line?

$$\hat{\text{Beer Price}} = 4.5594683 + 0.4514183 \times \text{Hot Dog Price}$$

b.

Test: $H_0: \beta_1 = 0$

$H_A: \beta_1 \neq 0$

Use $\alpha = 0.05$.

P-value = 0.1071

Reject or Fail to Reject H_0 ? Fail to Reject

Conclusion: *There is not enough evidence to conclude that Beer Price and Hot Dog Price have a non-zero slope.*

c. What percent of the variation in the beer prices is explained by the line?

$$r^2 \times 100\% \approx 8.7\%$$

d. Residual Analysis:

- Are there any problems with the residuals? If so, what are they and why are they problems?

yes, Residual by X Plot exhibits a funnel suggesting violation of the equal variance assumption.

- Does it appear the residuals are approximately normal? Explain.

yes, as shown by the Normal Probability Plot.

e. Interpret the slope value for this problem.

When Hot Dog Price increases by 1 unit, we expect the Beer price to increase as well by roughly 0.45 unit.

Example 2: Let's look at some data on Hollywood Movies

We will be examining if the Rotten Tomatoes Critic Score can predict the Rotten Tomatoes Audience Score

a. What should be the explanatory variable and what should be the response based on the goal above?

- $X = \text{Rotten Tomatoes Critic Score}$
- $y = \text{Rotten Tomatoes Audience Score}$

Linear Fit

$$\text{Audience Score} = 34.963505 + 0.5068554 * \text{Rotten Tomatoes Score}$$

Summary of Fit

RSquare	0.663174
RSquare Adj	0.6605
Root Mean Square Error	9.848784
Mean of Response	61.92188
Observations (or Sum Wgts)	128

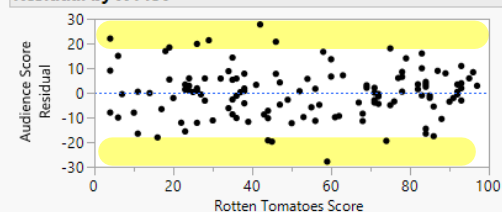
Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	24063.402	24063.4	248.0800
Error	126	12221.817	97.0	Prob > F
C. Total	127	36285.219		<.0001*

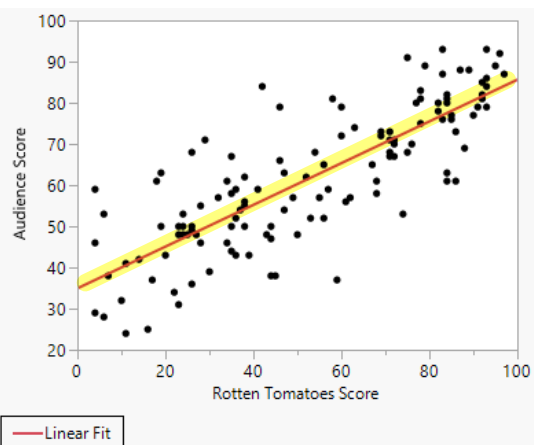
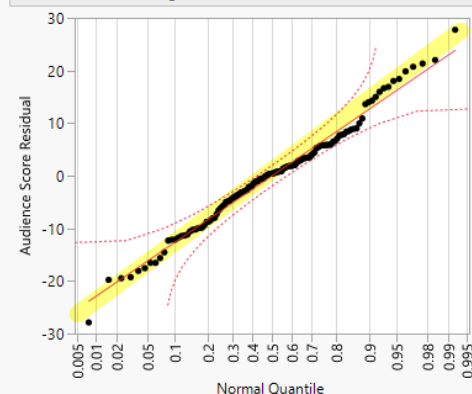
Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	34.963505	1.920238	18.21	<.0001*
Rotten Tomatoes Score	0.5068554	0.03218	15.75	<.0001*

Residual by X Plot



Residual Normal Quantile Plot



a. What is the equation of the least-squares line?

$$\hat{\text{Audience Score}} = 34.963505 + 0.5068554 \times \text{Rotten Tomatoes Score}$$

b.

Test: $H_o: \beta_1 = 0$
 $H_A: \beta_1 \neq 0$

Use $\alpha = 0.05$.

P-value = < 0.0001

Reject or Fail to Reject H_o ? Reject H_o .

Conclusion: There is evidence that Audience Score and Rotten Tomatoes Scores have non-zero slope.

c. What is the r-squared value?

$$r^2 \times 100\% \approx 66.3\%$$

d. Interpret the value you obtained in part (c) based on the definition.

Approximately 66.3% of the variability in Audience Score has been explained by the regression line with Rotten Tomatoes Score.

e. Is the line doing a good job? Justify your answer.

Yes, the line is doing a pretty good job in capturing not only the functional form but also much of the variation in the response variable.

f. Residual Analysis:

- Are there any problems with the residuals? If so, what are they and why are they problems?

No problem detected. The Residuals are randomly and consistently spread on the plot.

- Does it appear the residuals are approximately normal? Explain.

yes, as shown by the Normal Probability Plot.

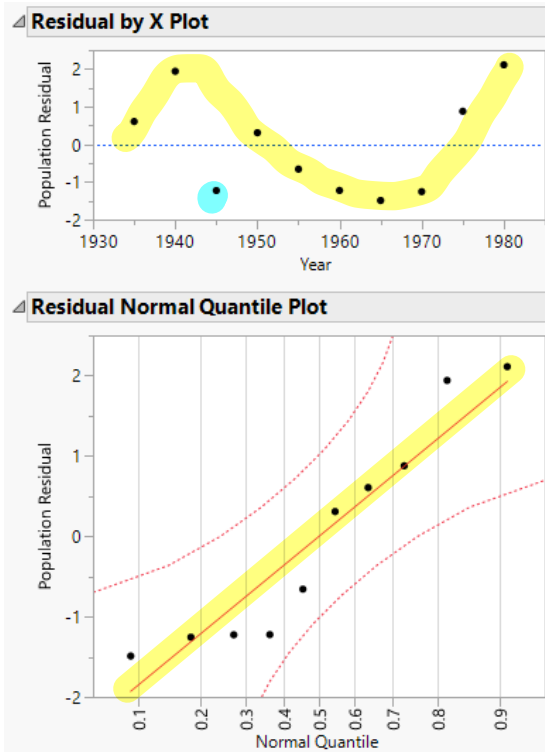
g. Interpret the slope value for this problem.

When Rotten Tomatoes score increases by 1 unit, then Audience score also increases by 0.5068554.

Example 3:

x = year

y = population on farms in U.S. (in millions)



Linear Fit

Population = 1166.9273 - 0.5867879 * Year

Summary of Fit

RSquare 0.977039
RSquare Adj 0.974169
Root Mean Square Error 1.444341
Mean of Response 18.29
Observations (or Sum Wgts) 10

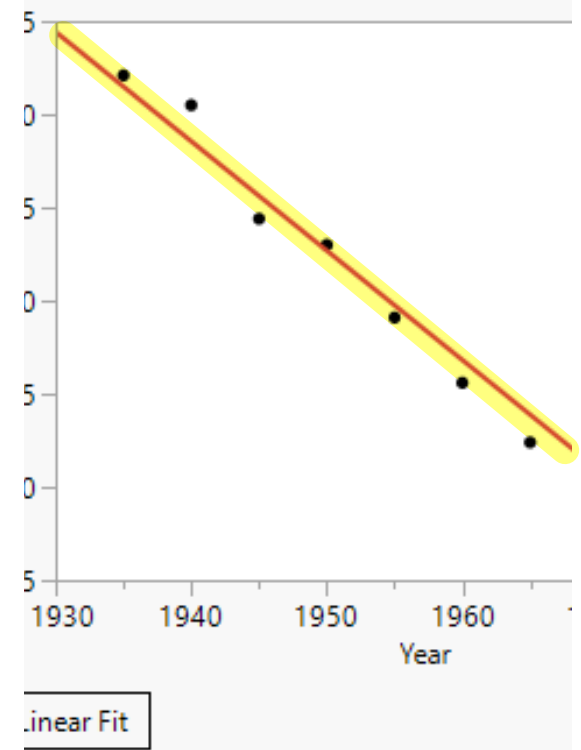
Analysis of Variance

Source	DF	Sum of Squares	Mean Square
Model	1	710.16003	710.160
Error	8	16.68897	2.086
C. Total	9	726.84900	

P-value < 0.0001

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob
Intercept	1166.9273	62.25672	18.74	<.0001
Year	-0.586788	0.031803	-18.45	<.0001



a. What is the equation of the least-squares line?

$$\hat{\text{Population}} = 1166.9273 - 0.5867879 \times \text{year}$$

b. Use the line to predict what the population on farms was in 1955.

$$\begin{aligned}\hat{\text{Population}} &= 1166.9273 - 0.5867879 \times (1955) \\ &\approx 19.75 \text{ millions}\end{aligned}$$

c. Use the line to predict what the population was on farms in 1981? How about what it will be in 2020?

$$\begin{aligned}\bullet \hat{\text{Population}} &= 1166.9273 - 0.5867879 \times (1981) \\ &\approx 4.5 \text{ millions}\end{aligned}$$

$$\begin{aligned}\bullet \hat{\text{Population}} &= 1166.9273 - 0.5867879 \times (2020) \\ &\approx -18.4 \text{ millions (Extrapolation Problem)}\end{aligned}$$

d.

Test: $H_o: \beta_1 = 0$

$H_A: \beta_1 \neq 0$

Use $\alpha = 0.05$.

P-value = < 0.0001

Reject or Fail to Reject H_o ? Reject H_o

Conclusion: *There is evidence that population and year have a non-zero slope.*

e. What is the r-squared value?

$$r^2 \times 100\% \simeq 97.70\%$$

f. Interpret the value you obtained in part (e).

About 97% of the population variance has been captured or explained by the regression

g. Is the line doing a good job? Justify your answer.

97% is very high and this suggests a strong line capturing most of the variability in population. Hence the line is doing a good job.

h. **Residual Analysis:**

Are there any problems with the residuals? If so, what are they and why are they problems?

The Residual by X Plot displays a cyclical pattern suggesting that a time series model would be a better fit to the data.

Does it appear the residuals are approximately normal? Explain.

yes. But since the sample size is small, we may not have an accurate guess of the distribution based on the normal probability.

i. Interpret the slope value for this problem.

From one year to another, the population is expected to increase by 0.5867879.



Reading

- Chapter 2 – Sections 2.2, 2.3
- Chapter 8 – Sections 8.1, 8.2



The End!

Thank you!